Natural Language Processing

Lecture 6: Applied NLP Tasks [Part II].

Main takeaways from previous lecture?

NER

Topic Modeling

POS

Summarization

Classification

Translation

Information Extraction

Question Answering

Document processing/OCR

Conversational

Applied NLP Tasks

POS

NER

Topic Modeling

Summarization

Classification

Translation

Information Extraction

Document processing/OCR

Conversational

Question Answering

There are no new models, there are new datasets

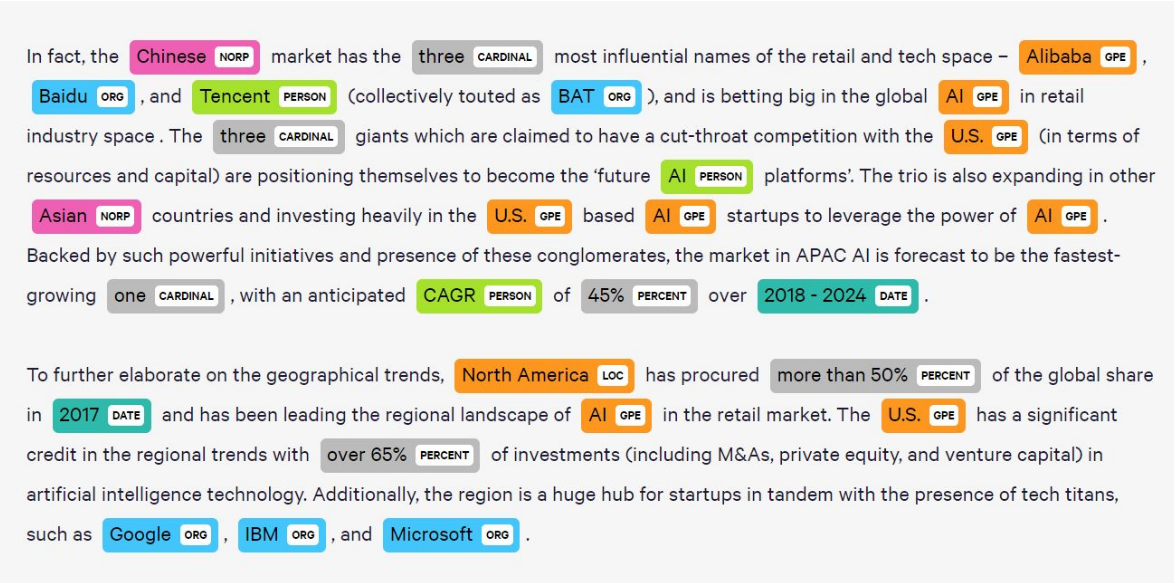
NER

Identifying and classifying named entities in text

Entities: people, organizations, locations, dates, etc.

Example: "Apple Inc. was founded by Steve Jobs in Cupertino in 1976"

- ORG: Apple Inc.
- PERSON: Steve Jobs
- LOC: Cupertino
- DATE: 1976



Why NER Matters

1. PII detection.
2. Information retrieval and question answering
3. Knowledge graph construction
4. Content recommendation
5. Medical text analysis (diseases, drugs, symptoms)
6. Financial analysis (companies, transactions)

Standard NER Categories

PERSON: individuals, fictional characters

ORGANIZATION: companies, agencies, institutions

LOCATION: cities, countries, geographic features

DATE: absolute or relative dates

TIME: times smaller than a day

MONEY: monetary values

PERCENT: percentages

NER Datasets

Biomedical: proteins, genes, diseases, drugs

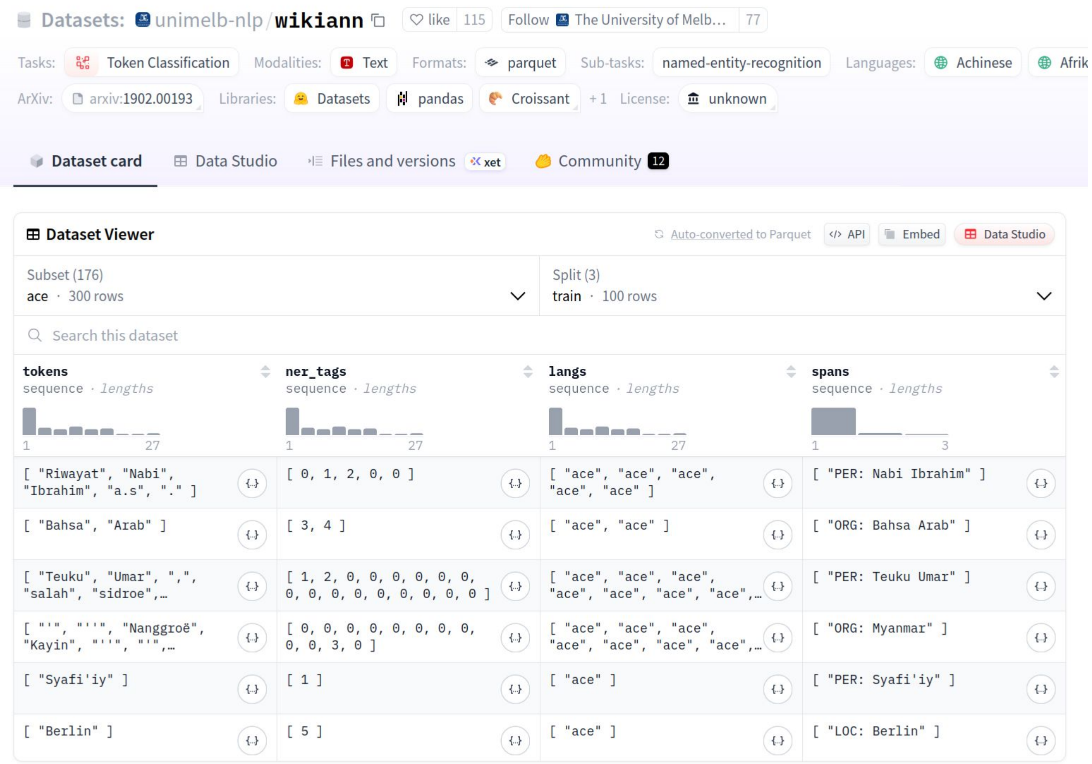
Legal: case numbers, statutes, court names

Scientific: chemical compounds, species names

Financial: stock symbols, financial instruments

Example (Medical): "Patient prescribed 50mg of As
for hypertension"

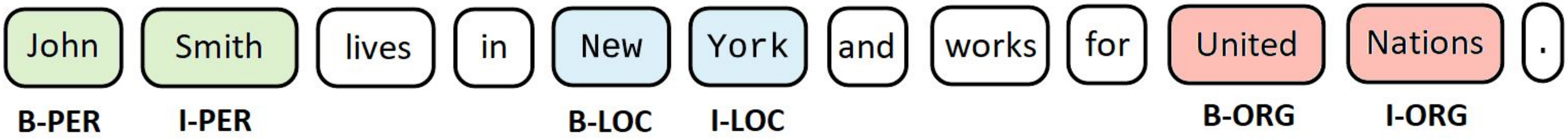
- DRUG: Aspirin
- DOSAGE: 50mg
- DISEASE: hypertension



Example dataset: WikiANN
https://huggingface.co/datasets/unimelb-nlp/wikiann

NER as Sequence Labeling

- Each token gets a label
- Input: ["Apple", "Inc.", "was", "founded", "in", "1976"]
- Output: [B-ORG, I-ORG, O, O, O, B-DATE]
- Sequence-to-sequence problem



Traditional NER: Rule-Based Systems

1. Hand-crafted patterns and dictionaries
2. Regular expressions for dates, emails, phone numbers
3. Gazetteer lists for locations, person names
4. Example rule: "[Capital Letter][lowercase]+ [Capital Letter][lowercase]+" for names

High precision, low recall

Traditional NER: Feature Engineering

1. Capitalization features (is_capitalized, all_caps)
2. Word shape (Xx, XXX, xx)
3. Prefix/suffix (Mr., Dr., -tion, -ing)
4. Part-of-speech tags
5. Word embeddings
6. Gazetteers (is in location list?)

Neural NER: Transformer-Based

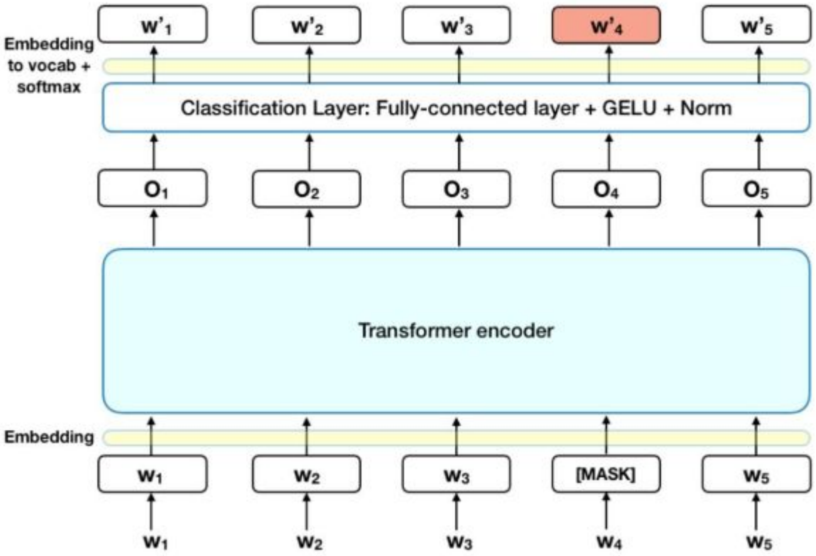
- BERT, RoBERTa, ELECTRA for NER
- Fine-tune pre-trained model on NER data
- Token classification head on top

Example: "Taylor Swift" → BERT → [B-PER, I-PER]

Current state-of-the-art

NER with BERT Example

- Input: "[CLS] Taylor Swift visited Paris [SEP]"
- BERT produces contextualized embeddings
- Classification layer on each token
- Output: [O, B-PER, I-PER, O, B-LOC, O]
- Fine-tune on labeled NER dataset



Zero-Shot and Few-Shot NER

Large language models!

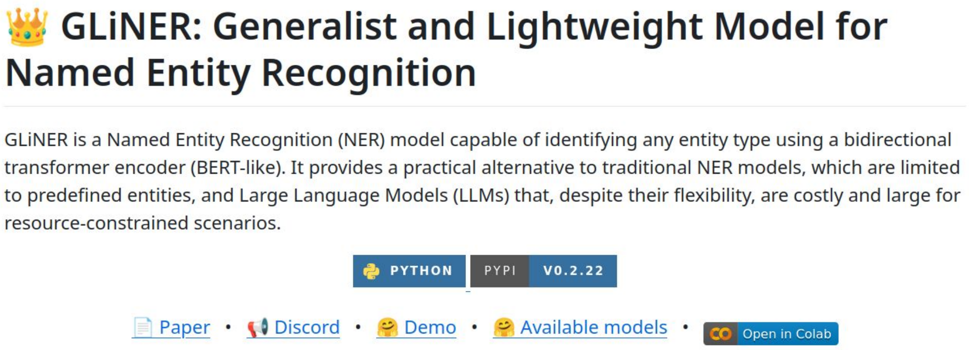
Prompt: "Extract person names from: 'Marie Curie won Nobel Prize'"

- Can specify entity types in natural language
- Useful when labeled data is scarce

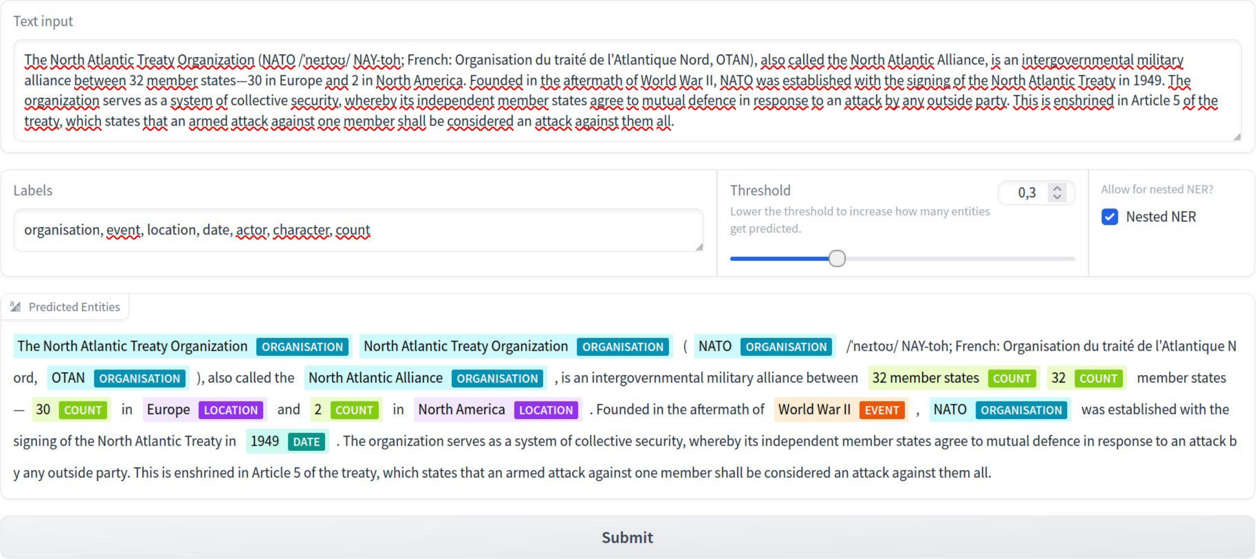
Disadvantages?

GLiNER models

- Can prompt to extract described entity
- Advantages of both LLMs and Encoder models
- Slightly less accurate than trained encoder model with classification head
- https://huggingface.co/spaces/urchade/gliner_multiv2.1



https://github.com/urchade/GLiNER



NER Challenges: Ambiguity

"Washington" - person, city, or state?

"Apple" - company or fruit?

Context crucial: "Washington crossed the Delaware" (PERSON)

"I visited Washington last week" (LOCATION)

NER Challenges: Boundary Detection

- Where does entity end?
- "Bank of America" vs. "Bank of America Corporation"
- "New York" vs. "New York City" vs. "New York Times"
- Nested entities: "University of [New York]"

Nested NER

- Entities within entities
- Example: "[Bank of [America]]" - LOC inside ORG
- "[University of [Southern California]]"
- Requires special model architectures
- Span-based or hypergraph-based approaches

NER Evaluation Metrics

- Precision: % of predicted entities that are correct
- Recall: % of actual entities found
- F1 score: harmonic mean of precision and recall
- Exact match vs. partial match
- Example: Gold: [B-PER, I-PER], Pred: [B-PER, O] → partial credit?

NER Practical Applications

- Customer service: extract account numbers, dates
- News analysis: track mentions of companies, people
- Healthcare: extract symptoms, medications from clinical notes
- Social media: detect trending locations, events

Part of Speech (POS) tagging

What is Part-of-Speech Tagging?

Assigning grammatical categories to words

Also called POS tagging or word-class disambiguation

Example: "They can fish"

- "They" (PRON) "can" (AUX) "fish" (VERB)

vs. "Fish can swim"

- "Fish" (NOUN) "can" (AUX) "swim" (VERB)

Universal POS Tags

NOUN: common nouns (dog, city)

PROPN: proper nouns (London, Mary)

VERB: verbs (run, eat)

ADJ: adjectives (big, red)

ADV: adverbs (quickly, very)

ADP: adpositions/prepositions (in, on)

CONJ: conjunctions (and, but)

DET: determiners (the, a)

PRON: pronouns (she, it)

Fine-Grained POS Tags: Penn Treebank

36 tags (more detailed than universal)

VB: verb base form (go)

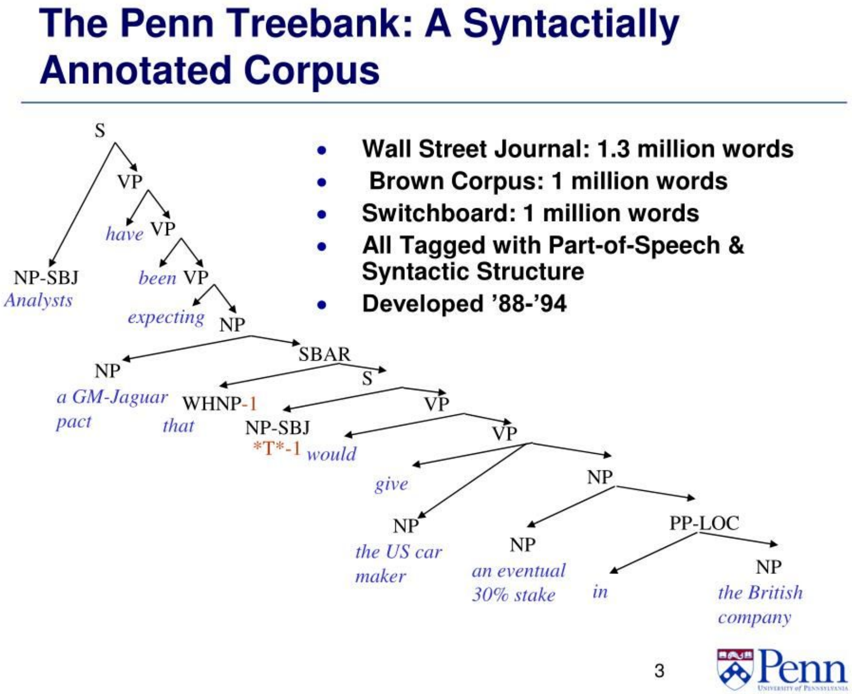
VBD: verb past tense (went)

VBG: verb gerund (going)

VBN: verb past participle (gone)

VBP: verb non-3rd person present (go)

VBZ: verb 3rd person present (goes)



POS Tagging Example

Sentence: "The quick brown fox jumps over the lazy dog"

Tags: [DET, ADJ, ADJ, NOUN, VERB, ADP, DET, ADJ, NOUN]

"The" (DET) "quick" (ADJ) "brown" (ADJ) "fox" (NOUN)

"jumps" (VERB) "over" (ADP) "the" (DET) "lazy" (ADJ) "dog" (NOUN)

Why POS Tagging Matters

- Preprocessing for parsing and NER.
- Information extraction (extract all nouns as entities).
- Machine translation (reordering based on grammar).
- Text-to-speech (pronunciation depends on POS).
- Query understanding ("book" as noun vs. verb).

Less relevant for most use cases now.

POS Tagging Ambiguity

Many words have multiple possible tags

"book": noun ("read a book") vs. verb ("book a flight")

"well": noun, adjective, adverb, interjection

"back": noun, verb, adjective, adverb

Context resolves ambiguity

Rule-Based POS Tagging

- Hand-written rules based on word forms and context
- "Words ending in -ing preceded by 'is/was' → VBG"
- "Words after 'the' → likely NOUN"
- Limited by exceptions and complexity

Example: "The running water" vs. "He is running"

Statistical POS Tagging: HMM

- Hidden Markov Models
- Hidden states: POS tags
- Observations: words
- Learn transition probabilities P(tag_i | tag_{i-1})
- Learn emission probabilities P(word | tag)
- Viterbi algorithm finds most likely tag sequence

HMM Example

P(DET→NOUN) = high, P(DET→VERB) = low

P("dog"|NOUN) = high, P("dog"|VERB) = very low

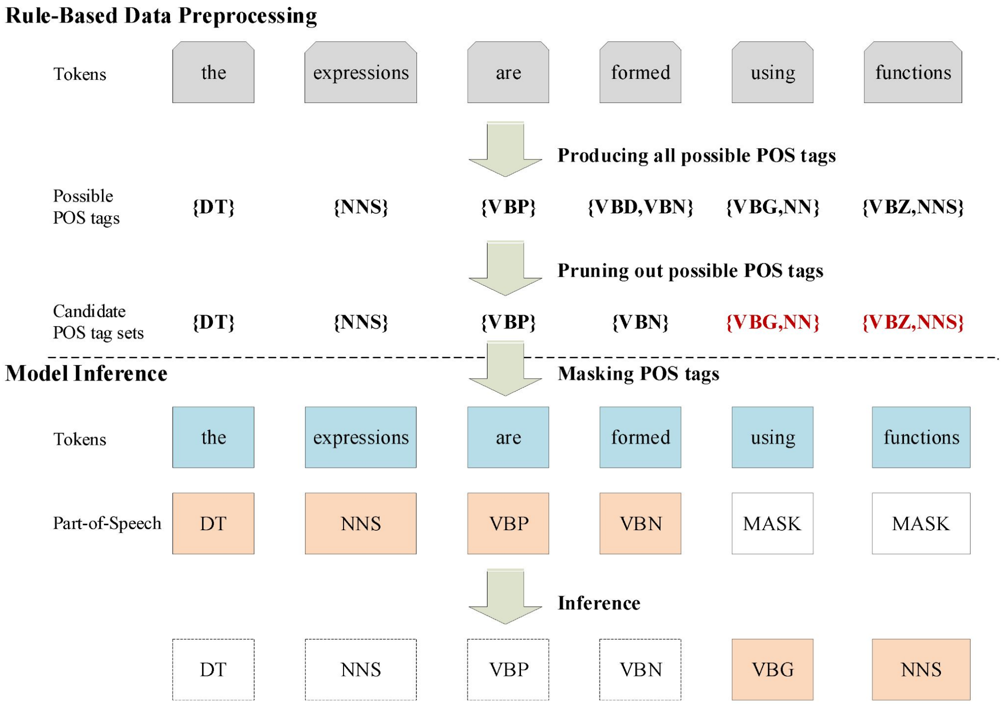
Sentence: "the dog runs"

Most likely: [DET, NOUN, VERB]

Not: [DET, VERB, VERB]

Transformer-Based POS Tagging

- BERT and other transformers
- Pre-trained contextual representations
- Fine-tune with POS tagging objective
- State-of-the-art accuracy (>97% on Penn Treebank)
- Fast inference with pre-computed embeddings



POS Tagging Evaluation

- Accuracy: % of correctly tagged tokens
- Typical accuracy: 95-97% for English
- Per-tag precision and recall
- Unknown word accuracy (important!)
- Example: 970 correct out of 1000 tokens = 97% accuracy

POS Tagging Challenges

- Unknown/rare words (proper nouns, technical terms)
- Cross-domain performance (trained on news, tested on tweets)
- Language-specific phenomena
- Informal text (social media, chat)
- Example: "gonna" → informal form of "going to"

Topic Modelling

What is Topic Modelling?

- Discovering abstract topics in document collections
- Unsupervised learning: no labeled data needed
- Each document is a mixture of topics
- Each topic is a distribution over words
- Example: news articles about "politics", "sports", "technology"

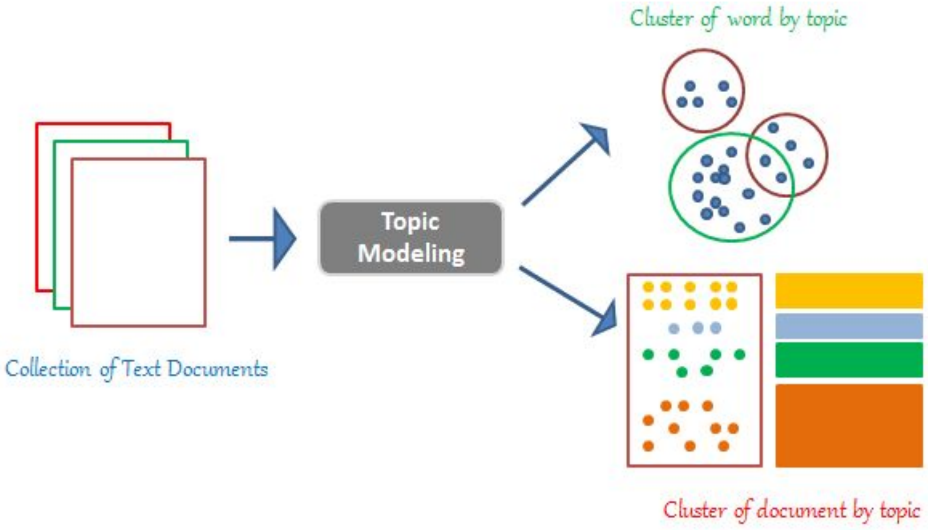
Topic Modelling Intuition

- Document: "The president announced new economic policies"

  Topics present: Politics (70%), Economics (30%)

- Document: "The team won the championship game"

  Topics present: Sports (95%), Entertainment (5%)



Applications of Topic Modelling

- Document organization and browsing
- Trend analysis over time
- Content recommendation
- Search result clustering
- Scientific literature analysis
- Social media monitoring

Latent Dirichlet Allocation (LDA)

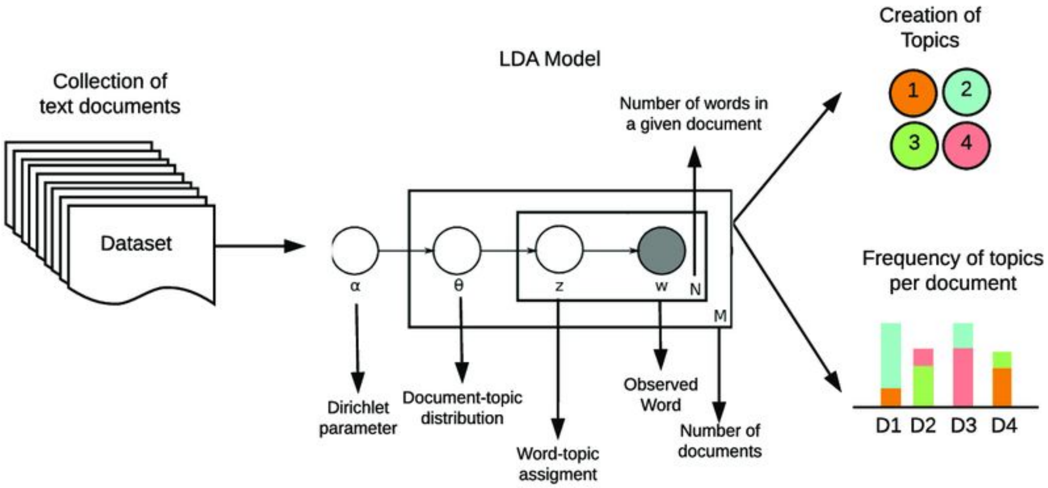
Most popular topic modelling algorithm

Generative probabilistic model

Assumptions:

- Documents are mixtures of topics
- Topics are mixtures of words

Parameters: number of topics K

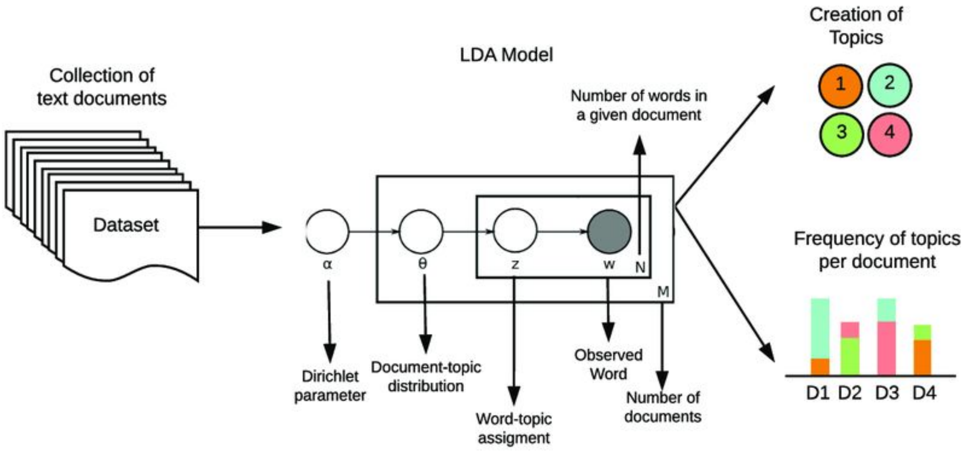


LDA Generative Process

For each document:

1. Choose topic distribution θ ~ Dirichlet(α)
2. For each word in document:
   - Choose topic z ~ Multinomial(θ)
   - Choose word w ~ Multinomial(φ_z)

Inference: reverse this process from observed words



LDA Example Topics

Topic 1 (Politics): government, president, election, policy, congress

Topic 2 (Sports): team, game, player, score, championship

Topic 3 (Technology): computer, software, data, algorithm, system

Each word has probability in each topic

LDA Document Representation

- Document D1: "New AI algorithm improves predictions"

  Topic distribution: Tech (80%), Science (15%), Business (5%)

- Document D2: "President signs new technology bill"

  Topic distribution: Politics (60%), Tech (30%), Law (10%)

Choosing Number of Topics

- Critical hyperparameter: K (number of topics)
- Too few: topics too broad, not useful
- Too many: topics too specific, redundant
- Methods: perplexity, coherence score, human evaluation
- Typically: 10-100 topics for moderate collections

Topic Coherence

- Measures interpretability of topics
- Do top words in topic co-occur in documents?
- High coherence: {car, vehicle, drive, wheel, engine}
- Low coherence: {car, happy, tree, government, eat}
- Automated metric but human judgment important

LDA Training

Expectation-Maximization or Variational Inference

Gibbs Sampling for posterior inference

Libraries: Gensim, Mallet,

Requires: preprocessing (stop words, stemming)

Iterative process until convergence

How to use

Training: scikit-learn.

Preprocessing for Topic Modelling

1. Remove stop words (the, is, at)
2. Lowercase conversion
3. Lemmatization or stemming
4. Remove rare and very common words
5. N-grams for phrases ("machine learning")

Example: "The dogs are running" → ["dog", "run"]

Topic Modelling Limitations

- Requires choosing K beforehand (except some variants)
- Topics may not align with human intuitions
- No semantic understanding (bag-of-words)
- Short documents problematic
- Doesn't capture word order

BERTopic

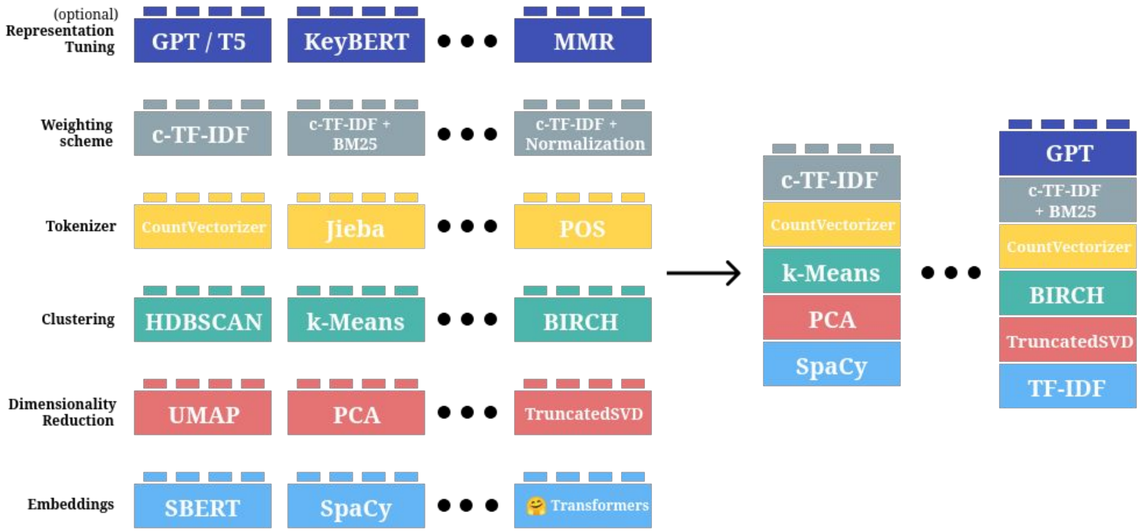
Modern approach using transformers

Steps:

1. BERT embeddings for documents
2. UMAP for dimensionality reduction
3. HDBSCAN for clustering
4. TF-IDF for topic representation

Better coherence, easier to interpret

Or use LLM for summarization to topic names?



Information Extraction

What is Information Extraction?

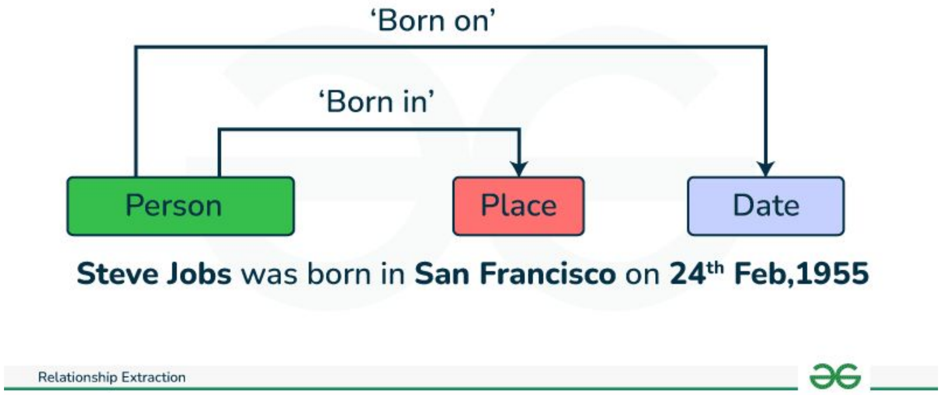
Extracting structured information from unstructured text

Beyond single entities to relationships and events

Goal: populate knowledge bases, databases

Example: "Tesla, led by Elon Musk, is headquartered in Austin"

Extract: (Tesla, CEO, Elon Musk), (Tesla, location, Austin)



Information Extraction Pipeline

1. Document preprocessing
2. Named Entity Recognition
3. Relation Extraction
4. Event Extraction
5. Coreference Resolution
6. Knowledge Base Population

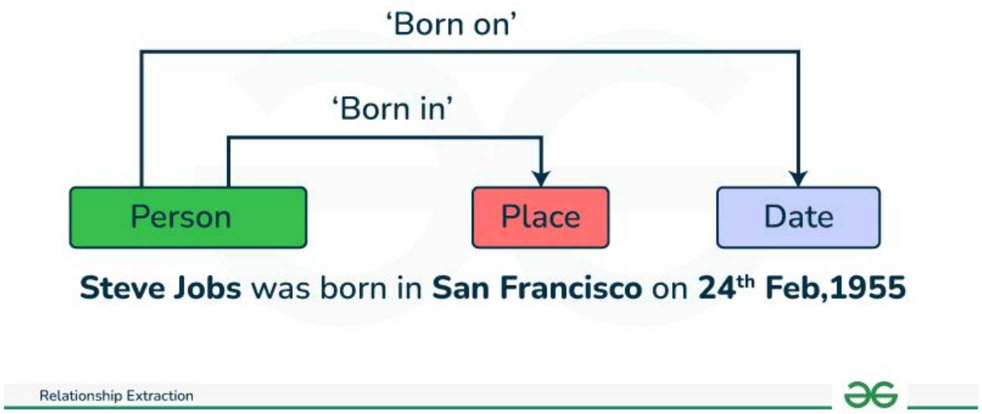
Relation Extraction

Identify semantic relationships between entities

Common relations: works_for, located_in, spouse, founded

Example: "[Steve Jobs] founded [Apple] in [1976]"

Relations: (Steve Jobs, founded, Apple), (founded, year, 1976)



Relation Extraction Approaches

Pattern-based: hand-crafted rules

- "X founded Y" → founder_of(X, Y)
- "X, CEO of Y" → works_for(X, Y)

Supervised learning: train classifier on labeled pairs

Distant supervision: use knowledge base to generate training data

Distant Supervision Example

Knowledge base: (Person 1, spouse, Person 2)

Find sentences containing both entities

"Person 1 and Person 2 attended the event"

Assume sentence expresses spouse relation

Generate training data automatically

Event Extraction

Identify events and their participants

Event: something that happens at a particular time/place

Example: "Apple acquired Beats for $3 billion in 2014"

Event type: Acquisition

Arguments: Acquirer (Apple), Target (Beats), Amount ($3B), Date (2014)

Event Extraction Structure

Trigger: word indicating event ("acquired")

Arguments: entities playing roles

Roles: buyer, seller, price, location, time

Example: "The company fired 100 employees last month"

- Trigger: "fired"
- Employer: company
- Employees: 100 employees
- Time: last month

Template-Based IE

Pre-defined templates for domains

Example template for job posting:

- Company: ___
- Position: ___
- Location: ___
- Salary: ___

Fill slots by pattern matching or ML

Open Information Extraction

Extract relations without pre-defined schema

Find all (subject, predicate, object) triples

Example: "Einstein developed the theory of relativity"

Extract: (Einstein, developed, theory of relativity)

No need to pre-specify relation types

Coreference Resolution

Determine which mentions refer to same entity

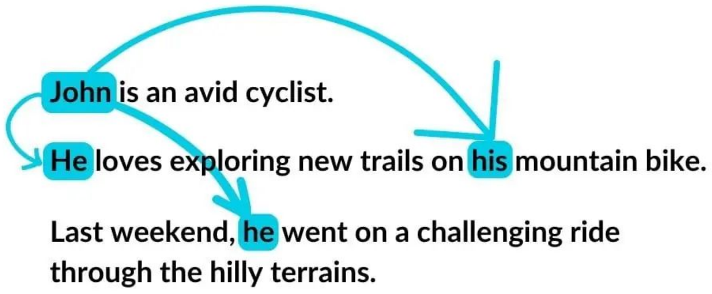
Crucial for IE accuracy

Example: "John met Mary. He greeted her warmly."

"He" → John, "her" → Mary

"The company announced... It will expand..."

"It" → The company



Coreference Challenges

Ambiguous pronouns: "The CEO told the manager he was wrong"

- Who was wrong? CEO or manager?

Semantic knowledge needed: "The composer wrote the symphony. He..."

- "He" likely refers to composer (male-dominated field traditionally)

Current systems use neural networks with attention

Knowledge Graph Construction

IE output → structured knowledge graph

Nodes: entities

Edges: relations

Example graph:

- (Apple) -[founded_by]-> (Steve Jobs)
- (Apple) -[headquartered_in]-> (Cupertino)
- (Steve Jobs) -[born_in]-> (1955)

IE Evaluation

Precision: % extracted facts that are correct

Recall: % of true facts that were extracted

F1 score: harmonic mean

End-to-end evaluation: from raw text to facts

Component evaluation: each stage separately

Document Processing

Document Understanding

- Processing documents beyond plain text
- Layout, structure, formatting matter
- PDFs, scanned documents, forms, tables
- Challenges: mixed content types, varied layouts

Relates to PDF processing as well

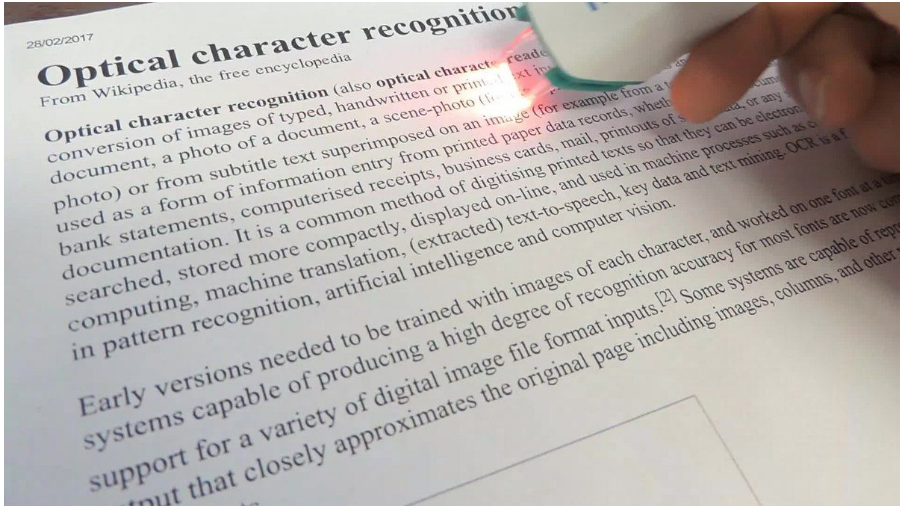
Optical Character Recognition (OCR)

Converting images of text to machine-encoded text

Scanned documents, photos of signs, historical documents

Two stages: detection (where is text?) + recognition (what is it?)

Applications: digitizing books, receipt scanning, license plates



OCR Pipeline

- Image preprocessing (binarization, noise removal)
- Layout analysis (find text regions)
- Text line and word segmentation
- Character recognition
- Post-processing (spell check, language model)

Isn’t it Computer Vision task?

Traditional OCR: Template Matching

- Match character images to templates
- Works for fixed fonts, clean images
- Fast but inflexible
- Fails on varied fonts, handwriting, degradation

Traditional OCR: Feature-Based

- Extract features (strokes, loops, endpoints)
- Classify using SVM, decision trees
- Tesseract OCR uses this approach
- Better generalization than template matching

Modern OCR: Deep Learning

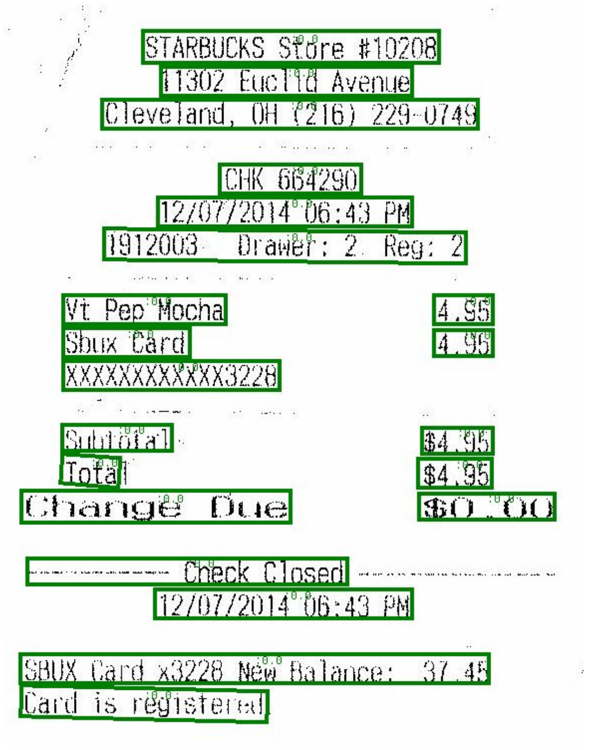
- CNN for character classification
- CRNN (CNN + RNN) for sequence recognition
- Attention mechanisms for focusing on characters
- End-to-end training from image to text
- State-of-art: Transformer-based (TrOCR)

OCR Example

Input: Image of receipt

Steps:

1. Detect text regions (store name, items, total)
2. Recognize characters in each region
3. Output: "Store: Walmart", "Item: Milk $3.99", "Total: $12.47"

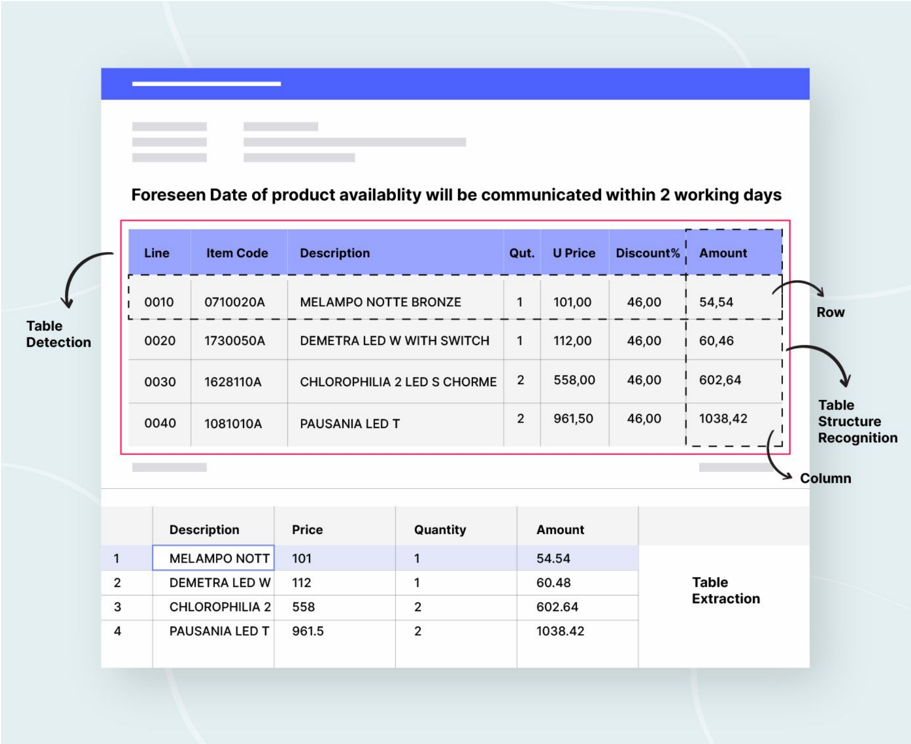


Document Layout Analysis

- Segment document into regions
- Headers, paragraphs, tables, figures, captions
- Important for understanding document structure
- Example: scientific paper → abstract, sections, references
- Methods: rule-based, ML classification of regions

Table Extraction

- Identify tables in documents
- Extract cell contents and structure
- Row/column relationships
- Challenges: spanning cells, nested tables, borderless tables, pivot tables
- Example: financial report with quarterly data table



Specialized networks, like TableTransformer from Microsoft.

Document AI Models

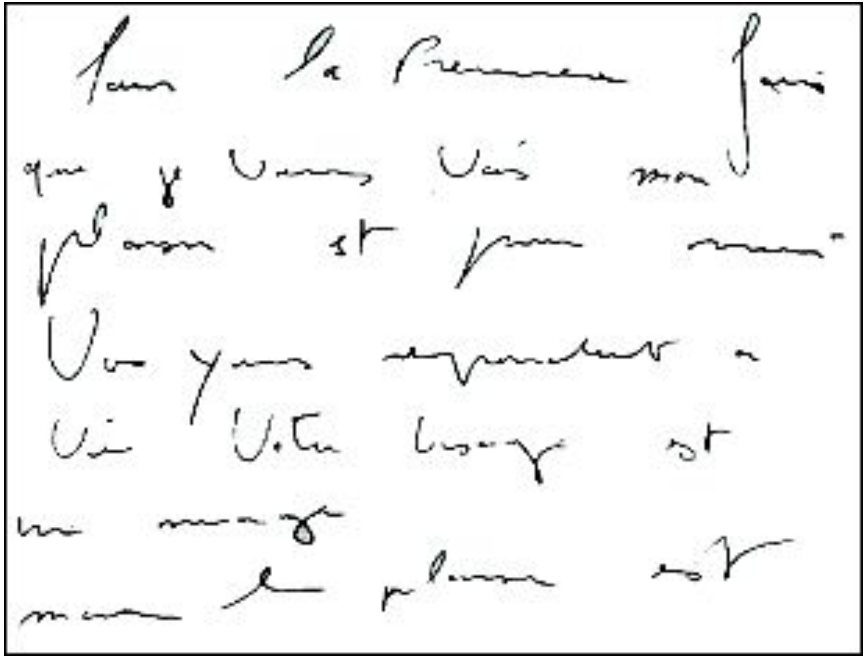
- LayoutLM: BERT for document understanding
- Combines text, layout, and image information
- Pre-trained on millions of documents
- Tasks: form understanding, receipt parsing, document classification
- Position embeddings for 2D layout

What to Use for Document OCR?

- Camelot
- Marker
- PyMuPDF

OCR Challenges

- Low quality images (blur, low resolution)
- Varied fonts and handwriting
- Multi-language documents
- Mathematical equations and symbols
- Complex layouts (newspapers, magazines)
- Historical documents (degradation, old fonts)



Summary and Next Steps

- NER: extracting entities from text
- POS: grammatical analysis
- Topic Modelling: discovering themes
- Information Extraction: structured data from text
- Document Processing: understanding visual documents

Q&A